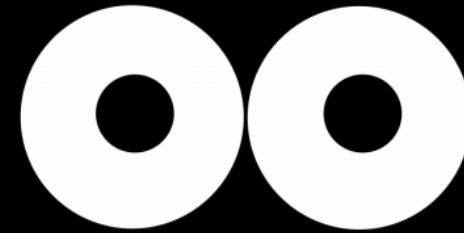


# Agentic AI Bootcamp Series

# #1

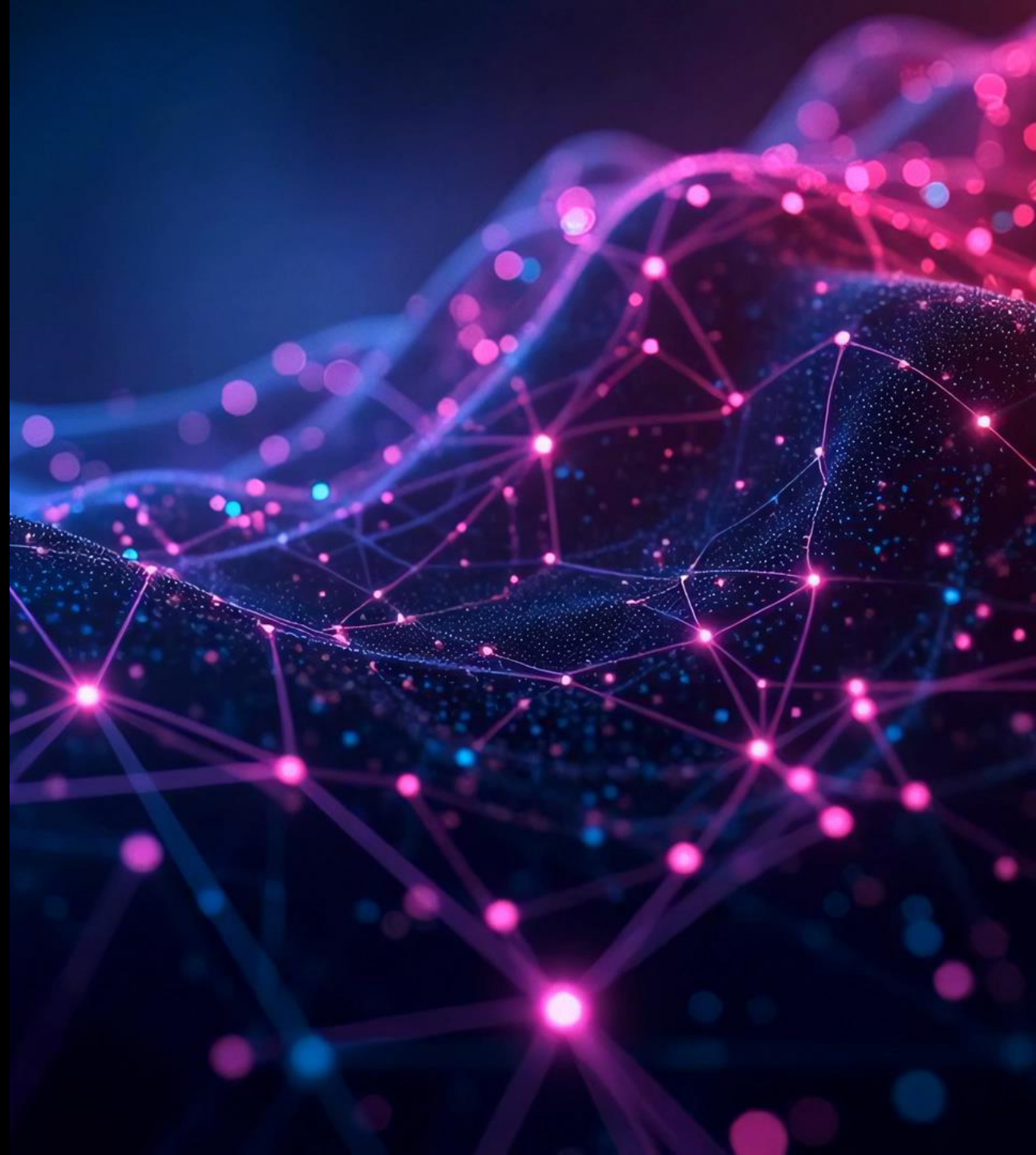
Agent demo are incredible, and this is exactly the issue.



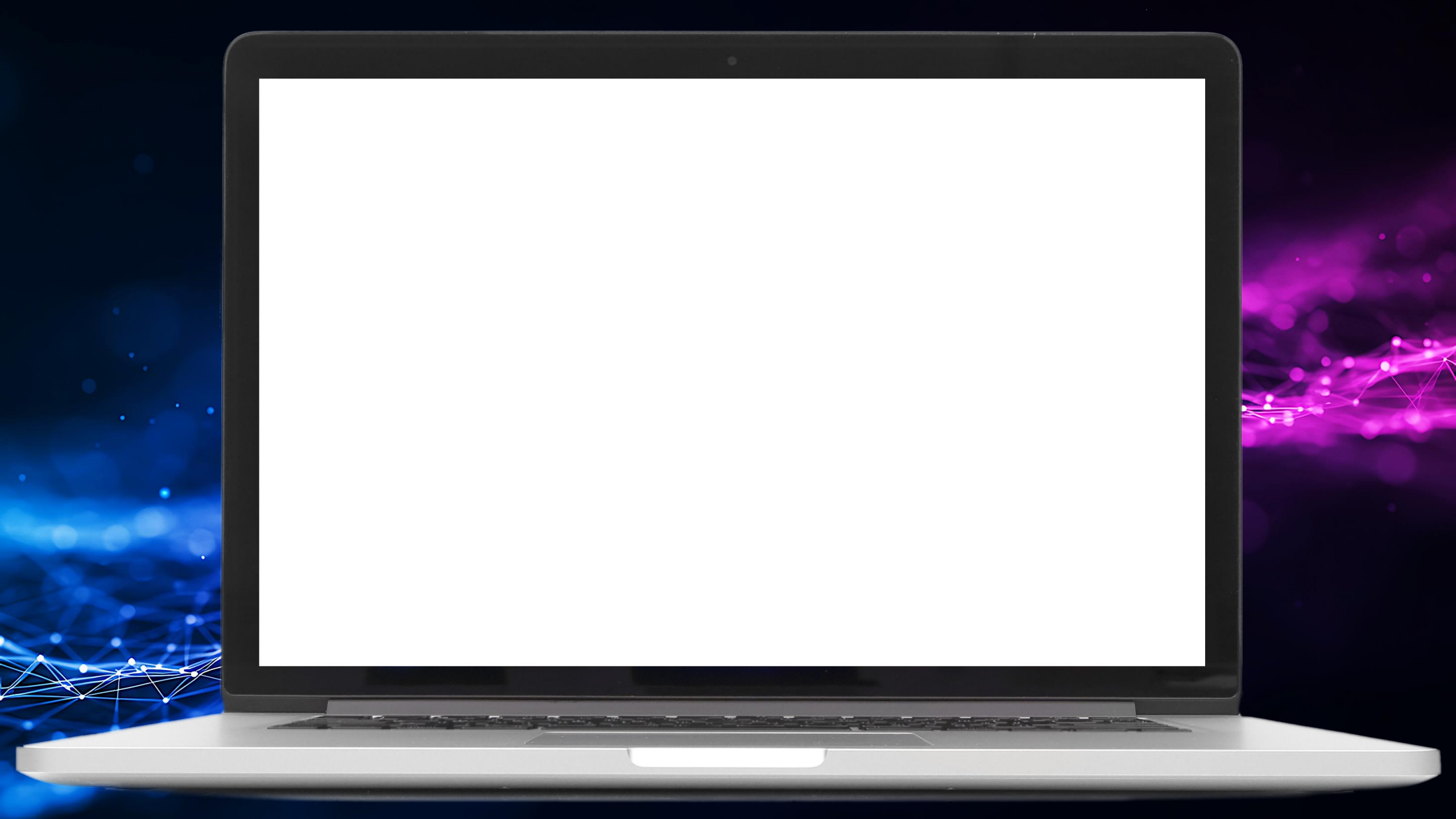
# Agenda

- Watsonx
- Agent Use case
- Why Agent
- How Agent works
- Agent Scale in Enterprise
- Custom Agent MCP – All together

→ Ready to BUILD





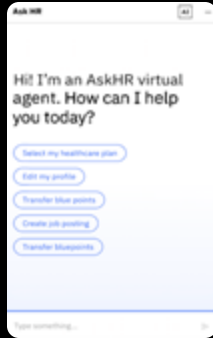




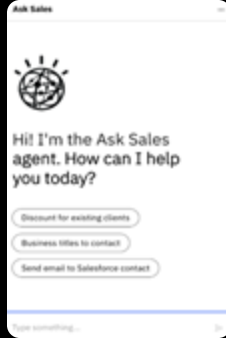
watsonx Orchestrate

## Prebuilt IBM's Agents

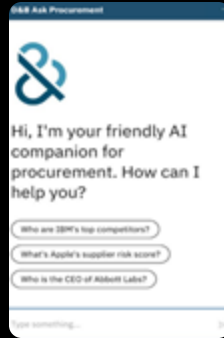
HR



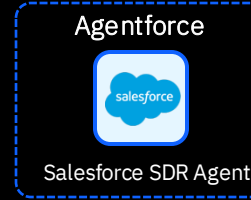
Sales



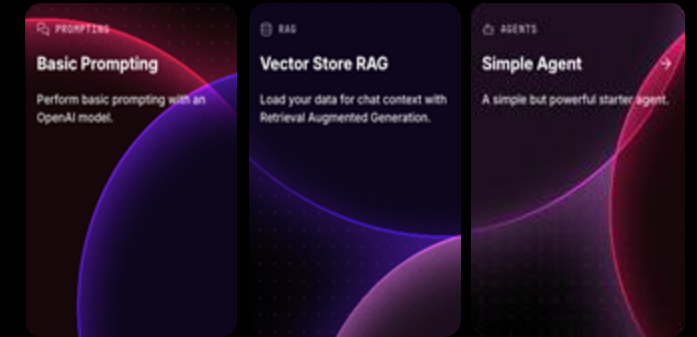
Procurement



## External Partner's Agents & Tools



## Build your own Agent



watsonx



Granite



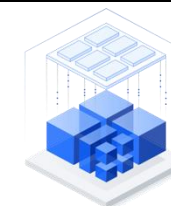
Meta



Hugging Face



OpenAI



IBM

DATASIX

# Use case

---

*Tal Neeman*

So, who went on vacation  
recently?

# But why Agents?

AI that can  
create for you



Models

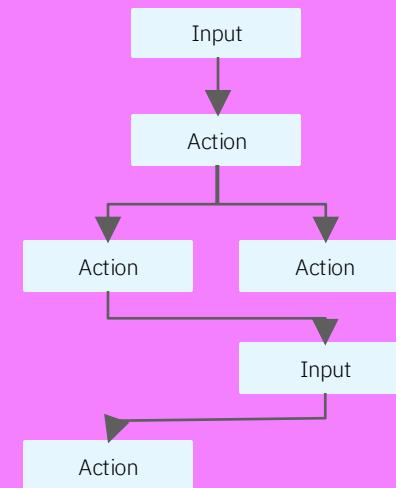
Problem-solving  
Logical thinking  
Pattern matching

AI that can  
chat for you



Assistants

Information retrieval  
Prescriptive tasks  
Single-step processes



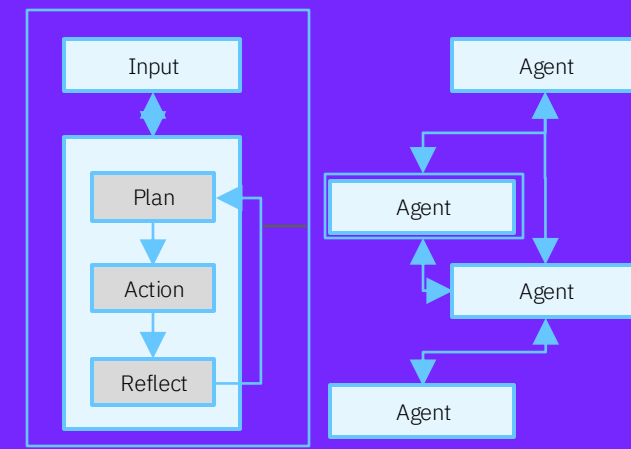
**Traditional assistants**

AI that can  
do for you



Agents

Multi-step processes  
Autonomous actions  
Self-corrections



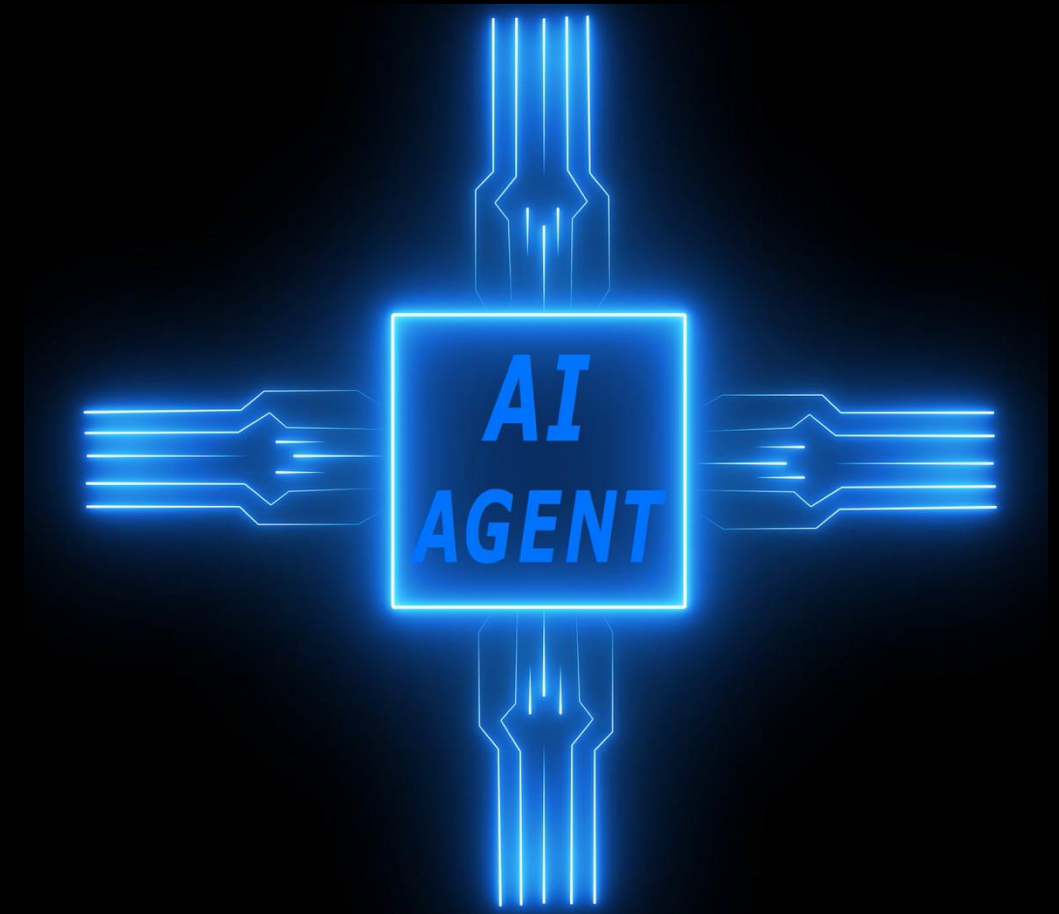
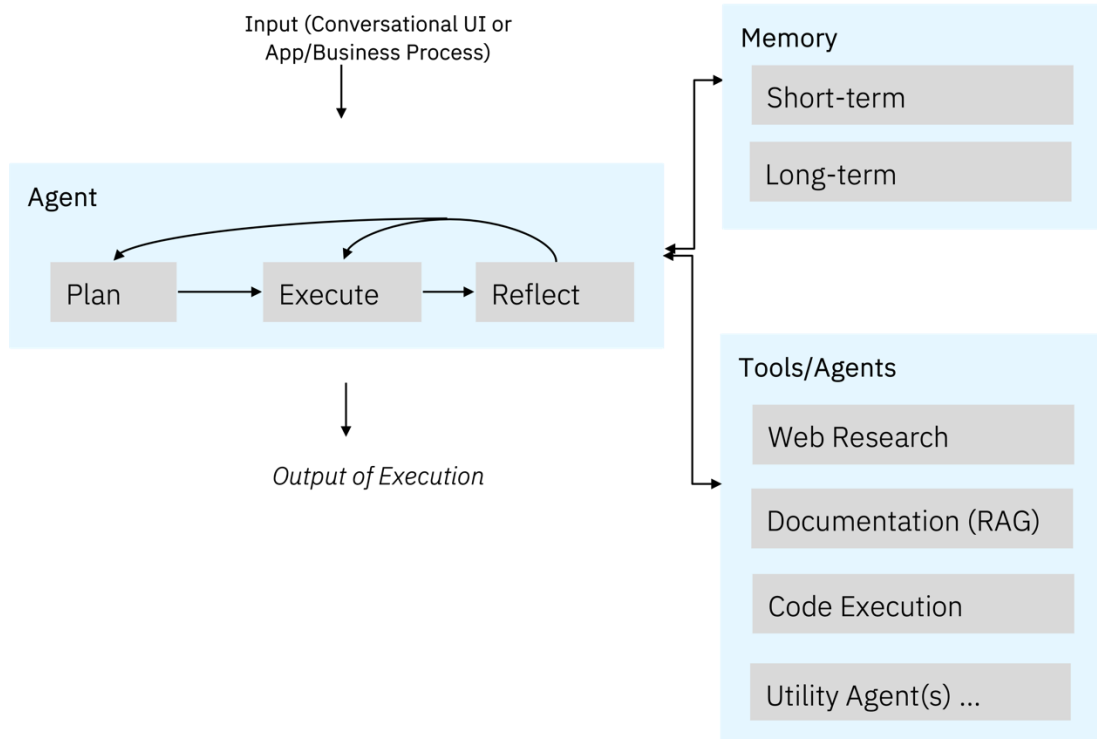
**Single-agent  
assistants**

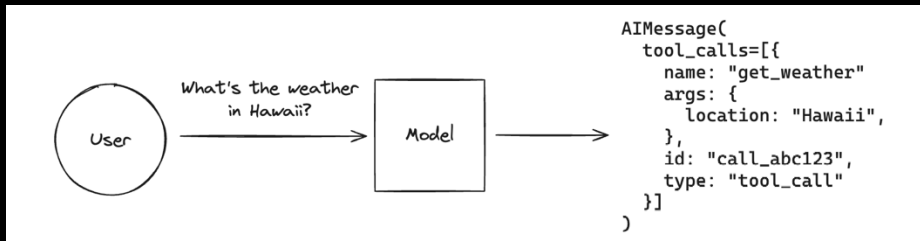
**Multi-agent  
assistants**



# AI Agents

*An AI agent is an autonomous system that can use tools and collaborate with other agents to plan and act on tasks. After it acts, the agent reflects on the results of its actions, learning iteratively and refining its approach to better align with its defined objectives.*





*Tool calling is just when an LLM writes a command to use a tool*

Hello Orchestrate! Can you please make me a market ... AI

You 06:46 PM

Hello Orchestrate! Can you please make me a market search for Datastax.

greeter 06:46 PM | Hide Reasoning ^

^ Step 1

Tool: langflow-mcp-toolkit:market\_research ← **Langflow MCP server tool for Market Search**

Input - 3 Lines

```
{
  "input_value": "Datastax"
}
```

Output - 3 Lines **Text**

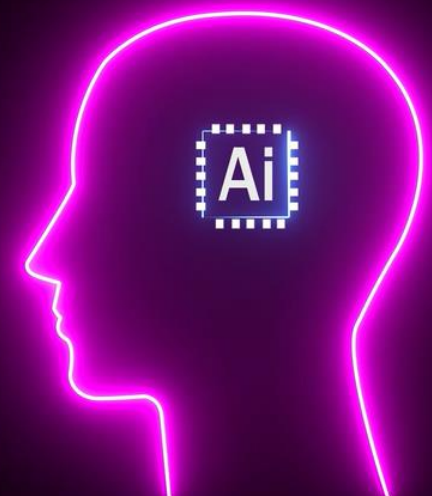
```
{
  "data": "meta=None content=[TextContent(type='text', text='Datastax', annotations=None), TextContent(type='text', text='\\n# Company Profile\\n\\n# Basic Information\\n- **Domain:** datastax.com\\n- **LinkedIn URL:**'

```

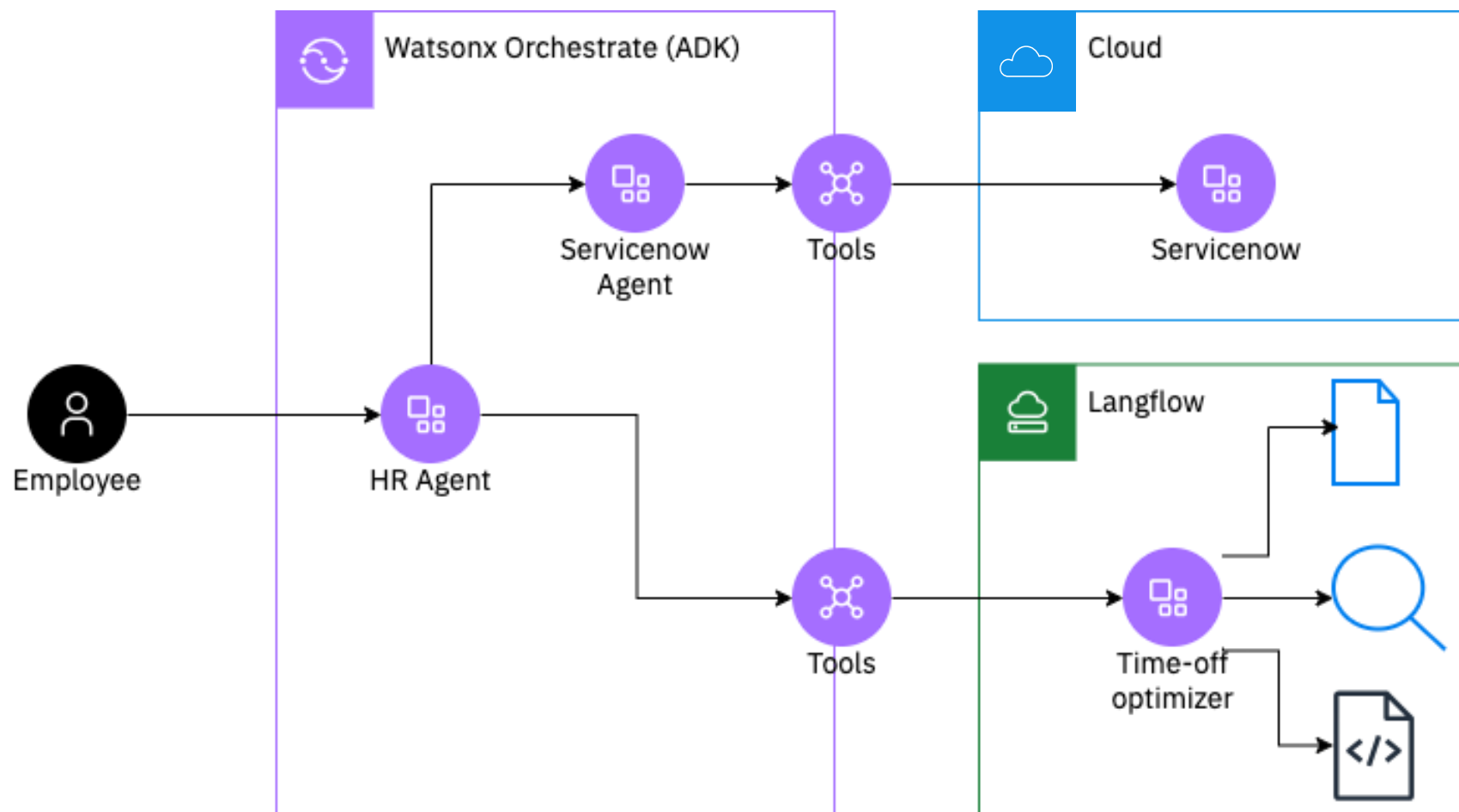
Show more v

Datastax is a company that provides an enterprise-grade database built on Apache Cassandra, with a cloud-native database-as-a-service (DataStax Astra) offering. Their key features include high availability and scalability, multi-cloud support, real-time data processing, advanced security features, analytics and search capabilities, developer tools, and graph database capabilities. Datastax targets industries such as finance, retail, telecommunications, and IoT, and offers subscription-based cloud services, support plans for on-premises deployments, and customized enterprise pricing.

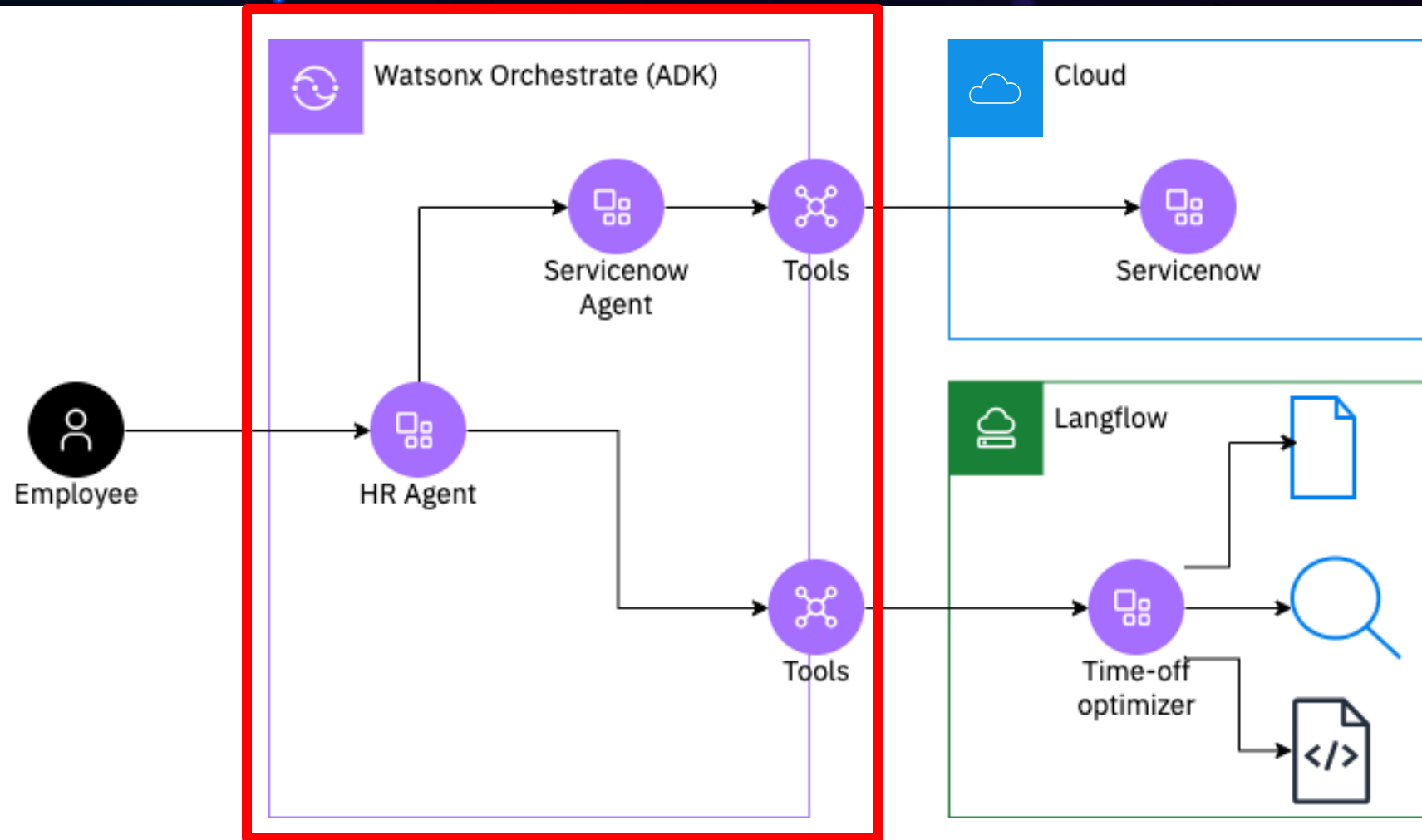
Type something... ➤



# Today's use-case: HR agents

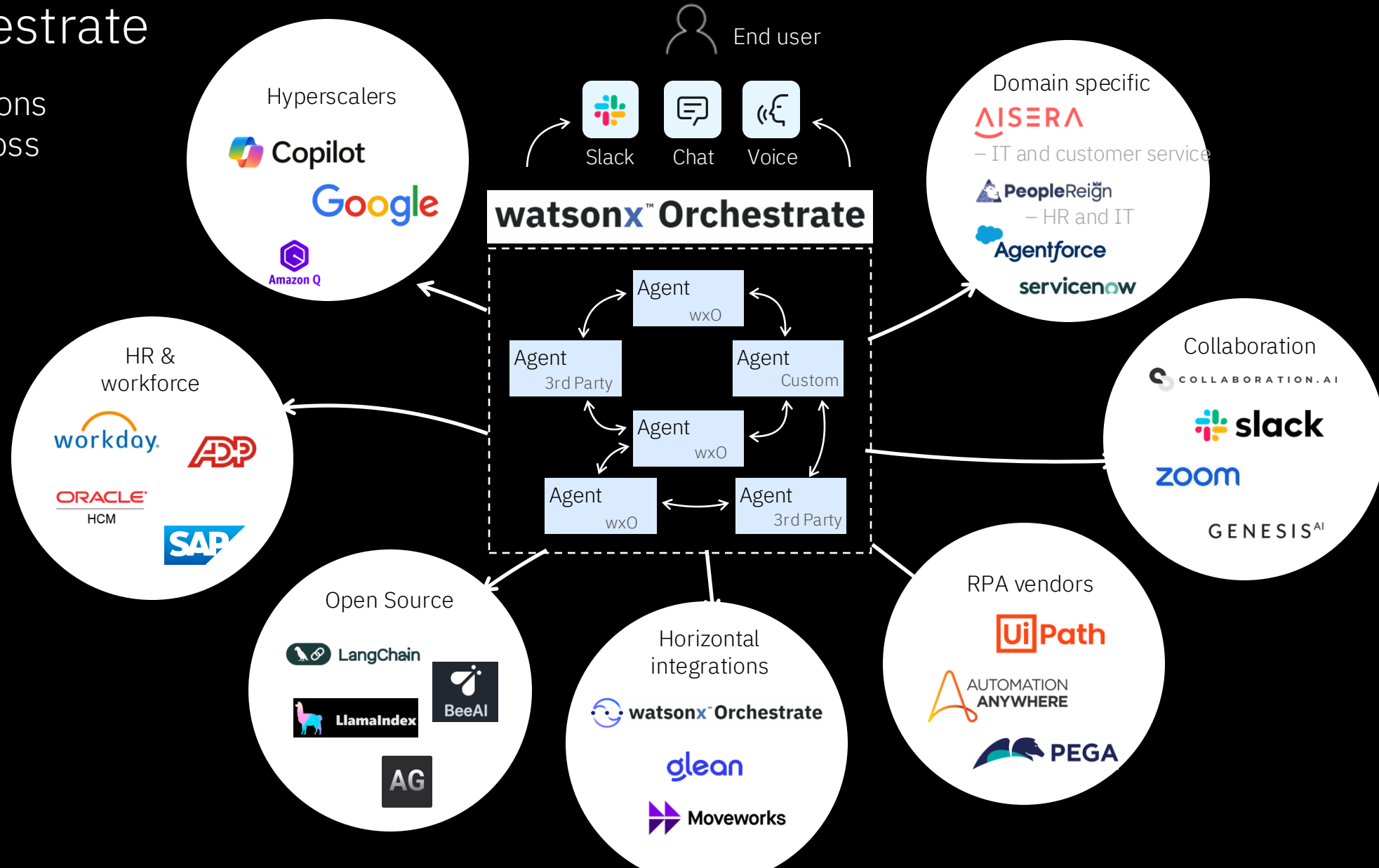


# Today's use-case: HR agents



# Managing the challenges of **Agentic Orchestration** with watsonx Orchestrate

Enabling ISV applications to manage agents across workflows

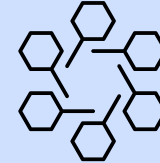




# Areas of AI Agent Innovation

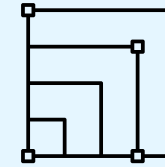
## Orchestrator for Tools and Agents

Multi-agent, multi-tool supervisor, router, and planner which facilitates complex task execution



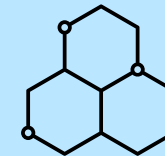
## Agent Marketplace and Prebuilt AI Agents

Accelerate AI Agents with pre-built utility agents and domain agents.  
Provide an easily searchable catalog of AI agents and tools.



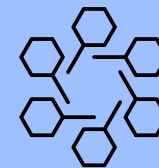
## Build Your Own Agents

Build custom designed agents with pro-code to no-code tooling  
Integrate 3<sup>rd</sup> party agents built in any tool or framework



## AI Agent Ops

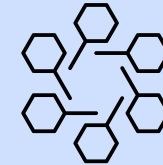
Discover, manage, monitor and optimize autonomous AI agents



# Areas of AI Agent Innovation

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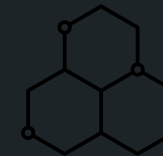
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Discover, manage, monitor and optimize autonomous AI agents

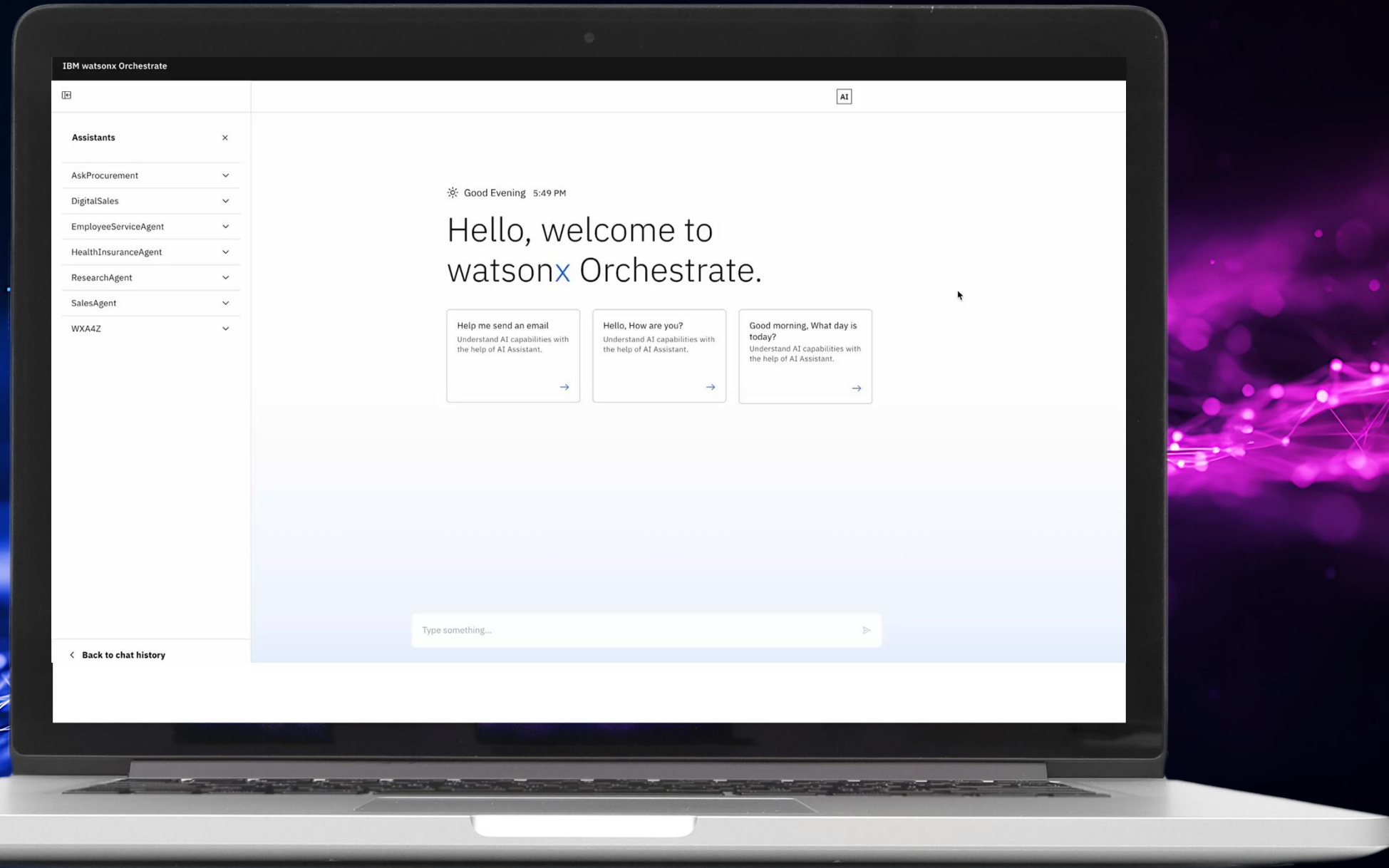


# Demo

## Recap: Multi-agent orchestration

Route between several  
Agents and Assistants

Builders can understand  
the chain of thought  
behind the LLM and how  
it routes and executes  
tasks



# Areas of AI Agent Innovation

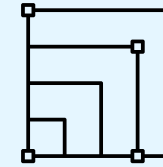
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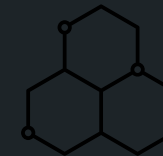
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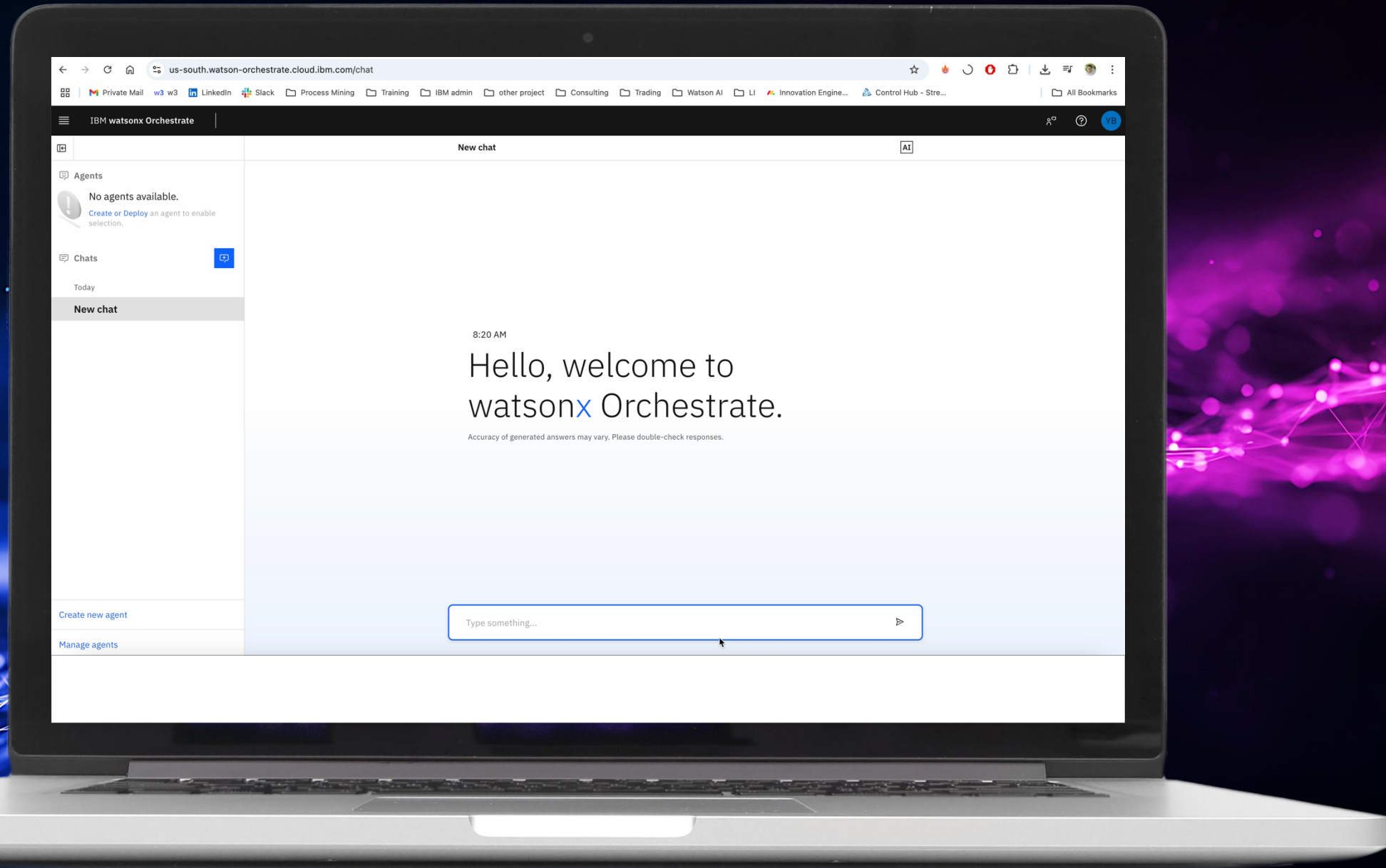
Discover, manage, monitor and optimize autonomous AI agents



# Demo

## IBM watsonx Agent Marketplace

An enterprise catalog of pre-built domain agents and tools for sharing, reuse, and distribution of 1<sup>st</sup> and 3<sup>rd</sup> agents



# Areas of AI Agent Innovation

## Orchestrator for Tools and Agents

Multi-agent, multi-tool supervisor, router, and planner which facilitates complex task execution



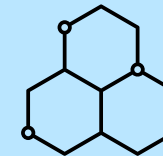
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## AI Agent Ops

Discover, manage, monitor and optimize autonomous AI agents





# watsonx Orchestrate no-code Agent Builder

The no-code AI Agent Builder redefines the wxO product experience:

- Build, deploy & manage AI agents intuitively in a guided flow, no-coding required
- Configure & deploy AI Agents grounded in business knowledge natively
- Connect directly to the tools & collaborator agents needed to compete workflows & reach outcomes
- Set agent behavior with instructions, rules & channels for control over the end-user experience
- Easily preview, deploy & manage agents with insights into agent use & edits

## Create an agent

### Create from scratch ☒

Build your custom agent step by step to create a custom solution.

### Create from template ☐

Browse the catalog and use the attributes of another agent as a template to create your agent.

### Profile

Describe your agent's purpose.

### Knowledge

Provide the specific knowledge your agent needs to accomplish its goal.

### Toolset

Add tools and other agents to help your agent take action.

### Behavior

Define how and where your agent should react to requests and respond to users.

# Demo Build AI agents with a low code ADK approach

Enterprise teams and developers can build expert agents with the agent ADK.

Full support for SDLC via GitOps and CI/CD integration

## Agents

Agents are used to orchestrate tools and other agents to perform complex tasks defined by the user. In Orchestrate, agents are available to users through a conversational interface, both through a UI client and through an API for direct integration. As Orchestrate is focused on empowering business teams, our agent framework is designed to be easy to use and understand for non-technical users and developers alike and supports both declarative and programmatic agent creation. The same components that are provided in the SDK are also used by the no-code Agent Builder UI (planned for public preview in Q2 2025).

Here is an example of an Agent:

```
# research_agent.py
from ibm_watsonx_orchestrate.agent_builder.agents import Agent
from ibm_watsonx_orchestrate.agent_builder.tools import tool
from .tools import web_search

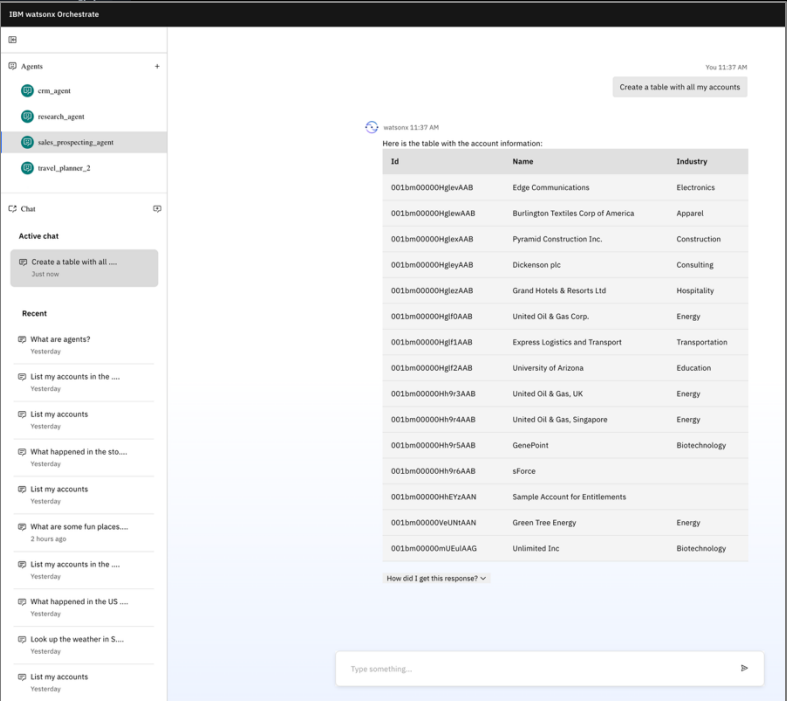
research_agent = Agent(
    name="research_agent",
    # The description will be used by Orchestrate to help it decide which Agent to use
    description="A Research Assistant which can query the web to help the user with research tasks that require research",
    instructions="Use the tools provided to answer the user's question. If you do not have enough information, use the web_search tool to find more information.",
    tools=[web_search],
)
```

They can also be defined in the equivalent YAML or JSON format:

```
# research_agent.yaml
spec_version: v1
kind: native
style: default
name: research_agent
description: A Research Assistant which can query the web to help the user with research tasks that require research
llm: watsonx/ibm/granite-3-2-8b-instruct
tools:
- web_search
```

The Orchestrate CLI can import an Expert Agents with the following command:

```
orchestrate agents import --file research_agent.yaml
```



# Areas of AI Agent Innovation

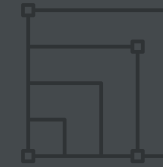
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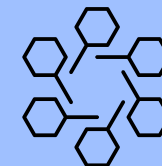
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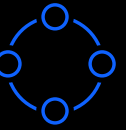


## AI Agent Ops

Discover, manage, monitor and optimize autonomous AI agents



# AI Agent Ops



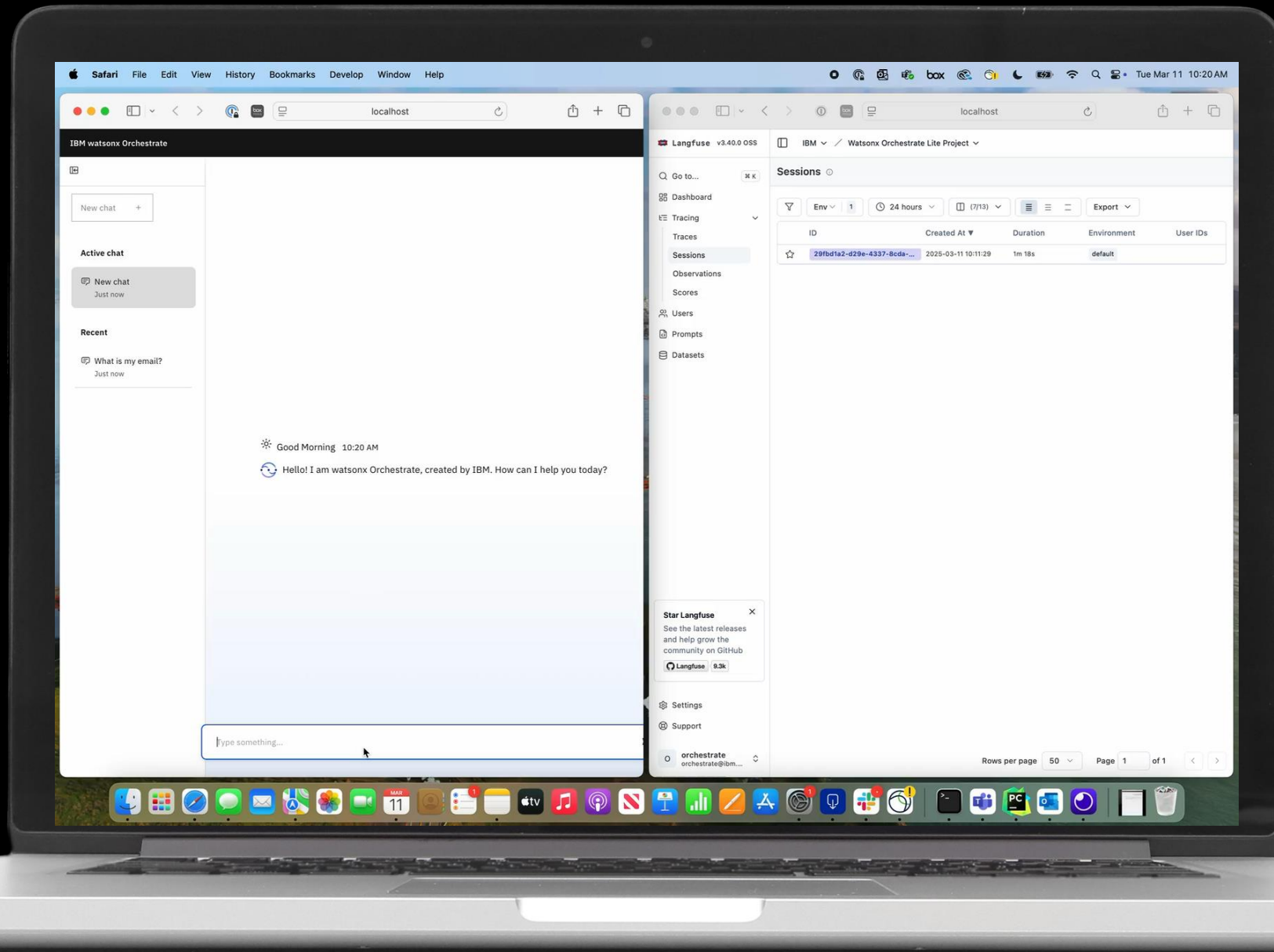
Real time observability:  
visualize end to end  
multi-step Agents  
operation flow

## Evaluation:

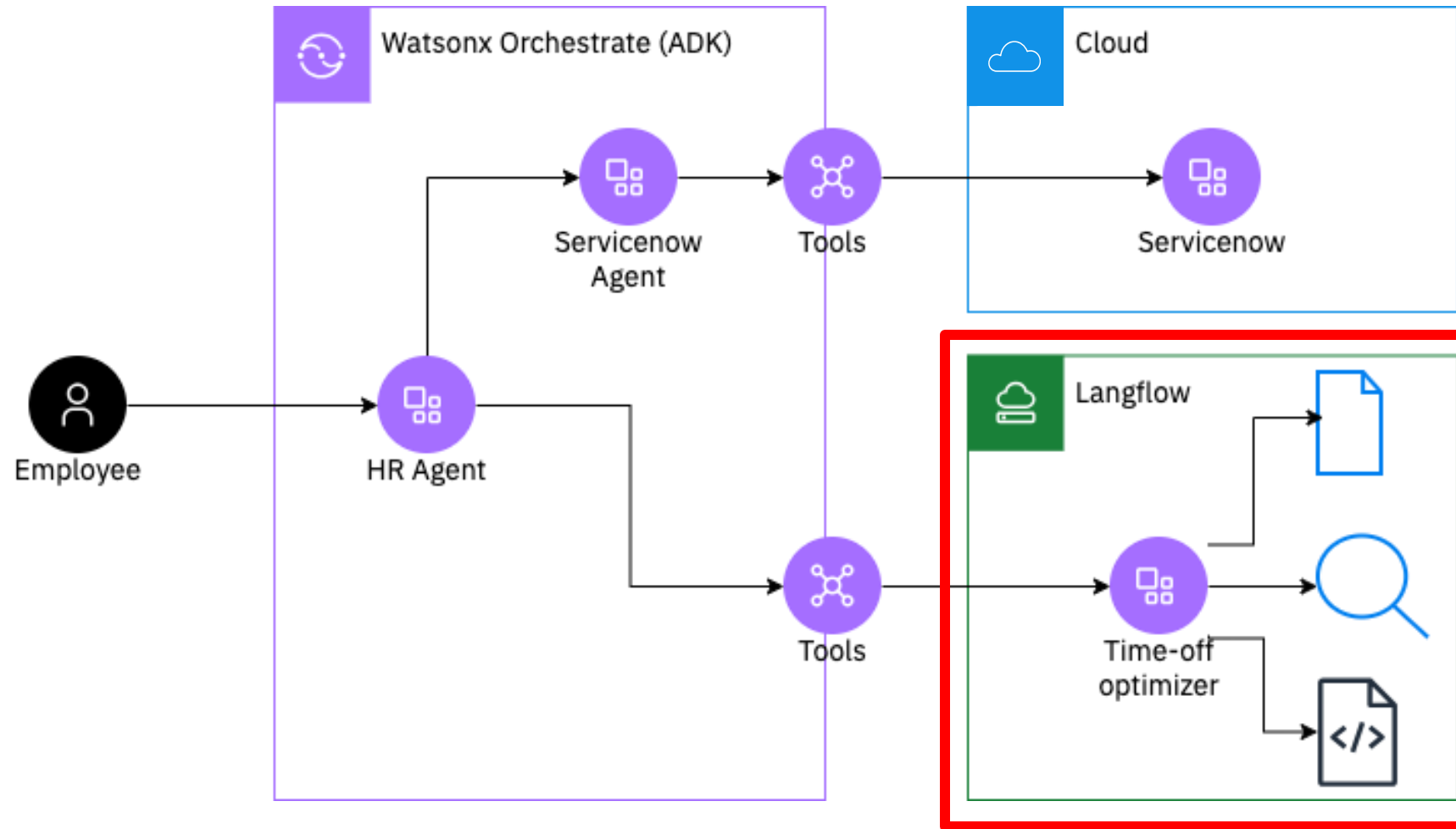
Governance framework  
for RAG tool & non-RAG  
tools evaluation for  
answer correctness,  
relevance

## Analytics:

Monitor the Agents  
performance, utilization,  
improve effectiveness,  
Business KPIs fulfillment



# Today's use-case: HR agents



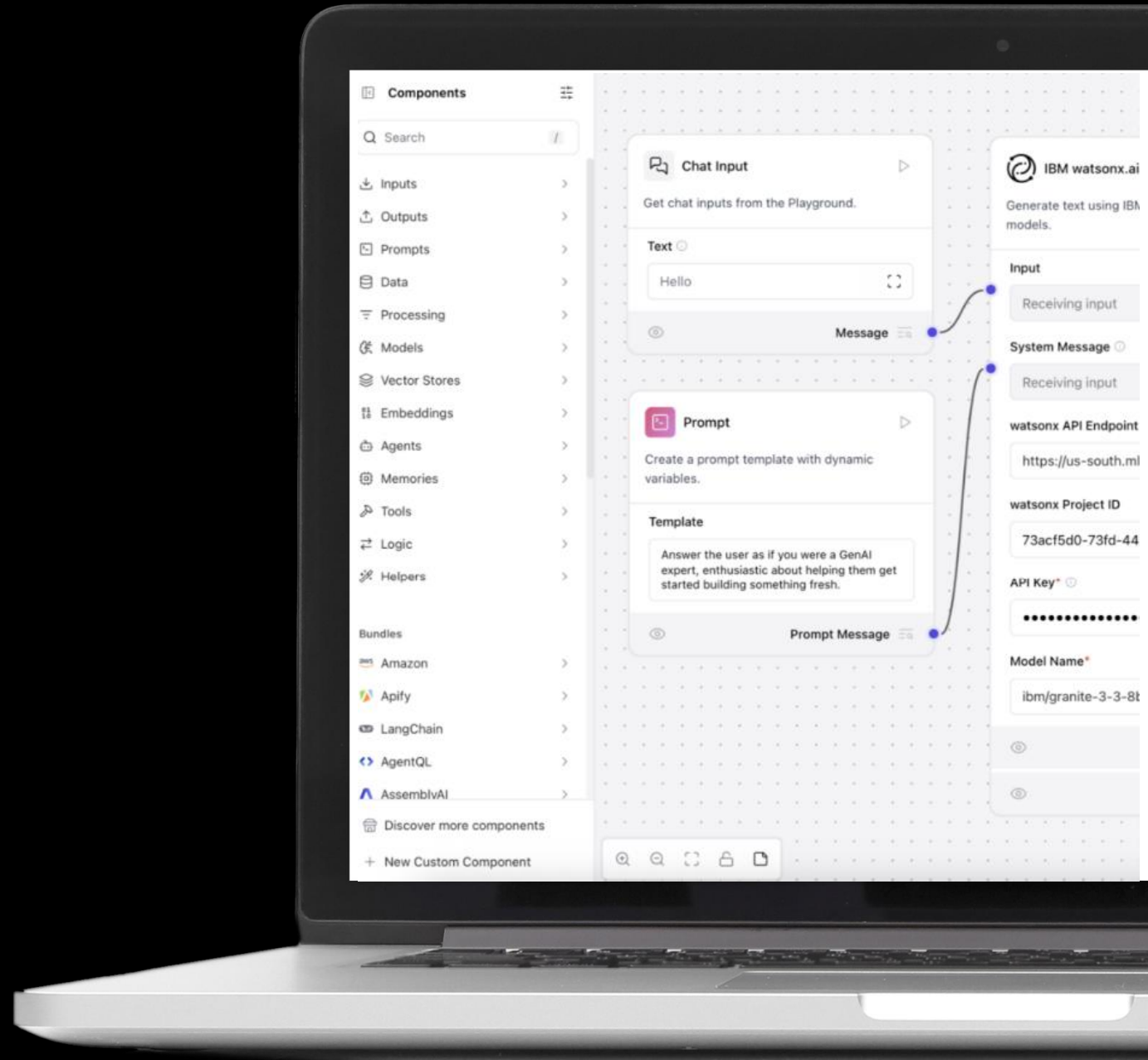
# What is Langflow?

- open-source platform for building AI flows through a visual interface.

Drag-and-drop components like language models, prompts, databases, and tools onto a canvas and connect them to create.

Langflow comes with a large library of pre-built components for common AI tasks, often referred to as the largest agent ecosystem in the low-code AI space.

Trusted by the community with over **60,000 GitHub stars**, showing strong adoption and support.





My Projects / Watsonx Agent ▼ Saved

Star 58,570

Components

Data

URL

Bundles

Youtube

Youtube Playlist

Discover more components

+ New Custom Component

Message

Language Model

Agent

Define the agent's instructions, then enter a task to complete using tools.

Language Model

Select an agent llm

Agent Instructions

You are a helpful assistant that can

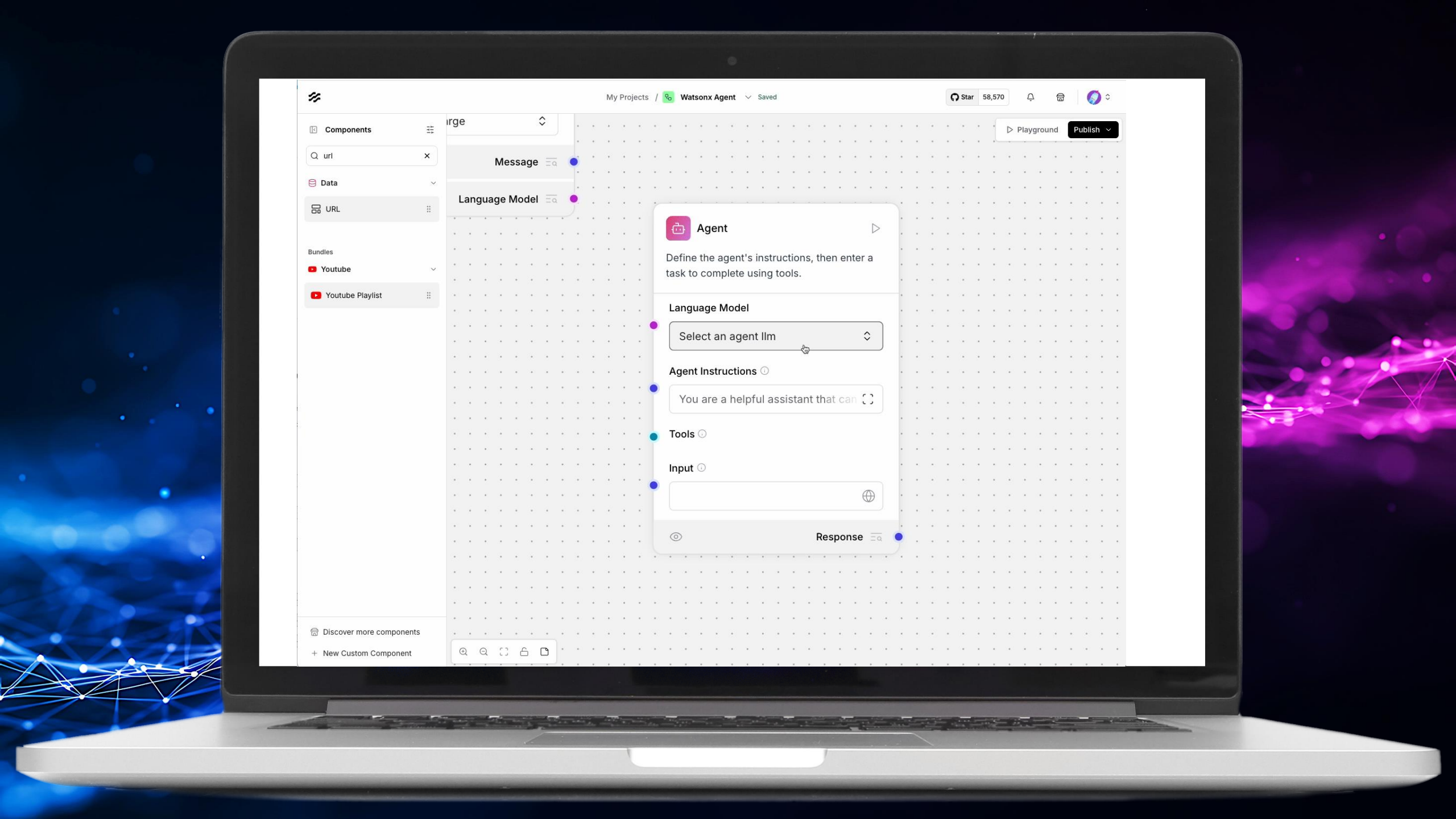
Tools

Input

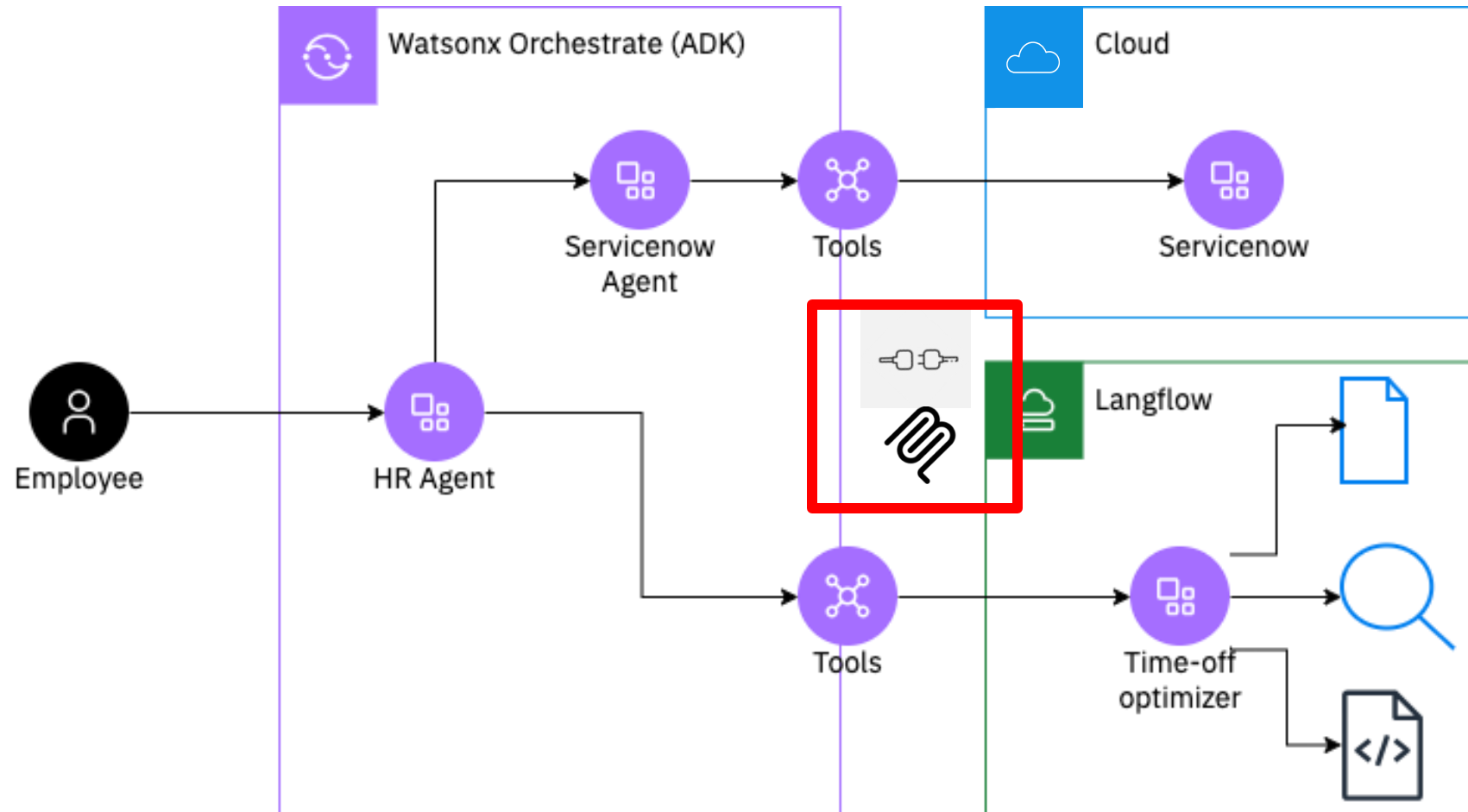
Response

Playground

Publish



# Today's use-case: HR agents



# MCP Architecture (Model Context Protocol - MCP)

1. **MCP Host** – The AI model requests data or performs any action.

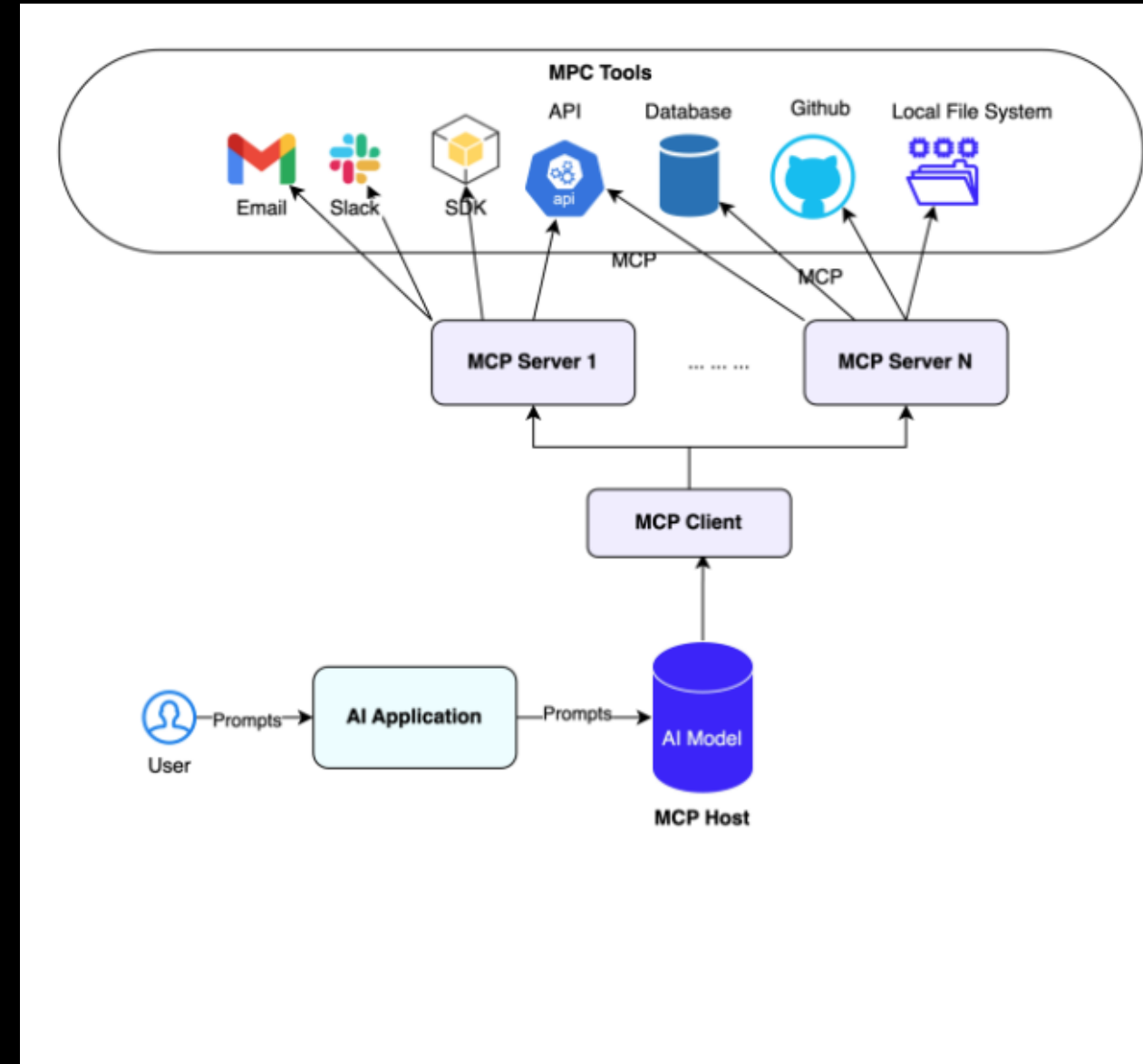
2. **MCP Client** – A client that forwards the AI model's requests to MCP servers.

- Invokes Tools
- Queries for Resources
- Interpolates Prompts

3. **MCP Server** – Lightweight applications that expose specific capabilities (APIs, databases, files, etc.) through standardization. It can be local or **Steamtable HTTP** (remote). Expose Tools, Resources and Prompts

4. **Data Sources** – Various backend systems, including local storage, file system, cloud databases, and external APIs. It can be local or remote.

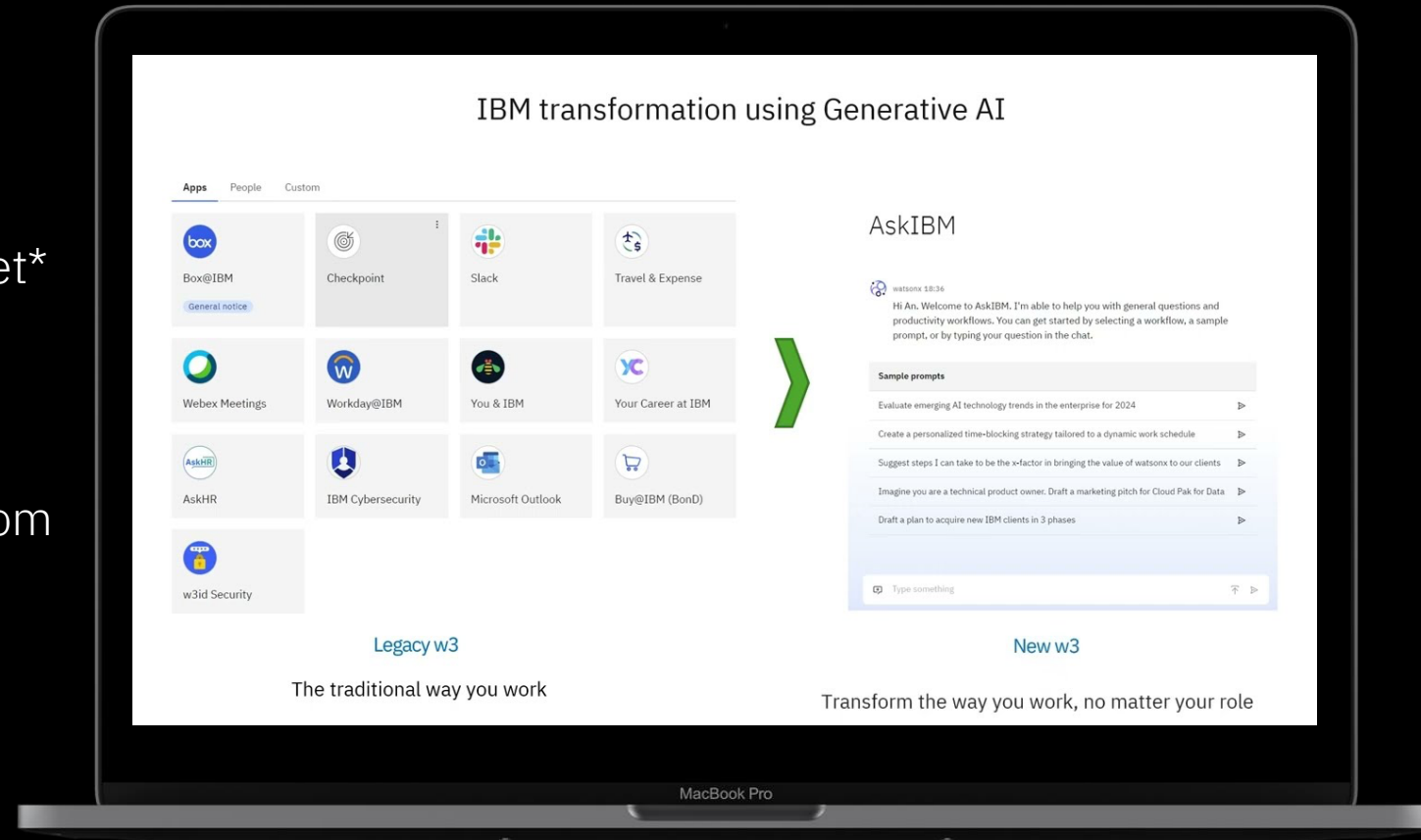
- **Tools:** Model Controlled Functions that Retrieve, Send or Update
- **Resources:** Local files, Database Records, etc.
- **Prompts:** Pre-defined template for AI interactions



# IBM is Client Zero for AI driven enterprise innovation

## IBM AskHR

- 10.1M Yearly interactions
- 765K Automated Tasks
- 40% Reduction in HR Operating Budget\*
- 94% Employee inquiries handled by AskHR
- +55 Improvement of HR NPS score from +19 → +74
- 96% Manager adoption
- 93% Executive adoption
- 2 Bands Average increase in HR 'level'



Build Travel Agent  
Langflow

with

<https://ibm.biz/langflow>

Meal



# Build HR Agent with Orchestrate

# Build Agent & Connect



What's on AI

Groupe WhatsApp



Join the  
**watsonx** Israel  
Community